9.9.5 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered …

One mistake I learned from today …

One thing I am not sure about … YET!

One thing I am confident I know is …

Unit 9, Lesson 10: Applying Different Methods of Solving a Quadratic

Benchmarks

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

Learning Targets

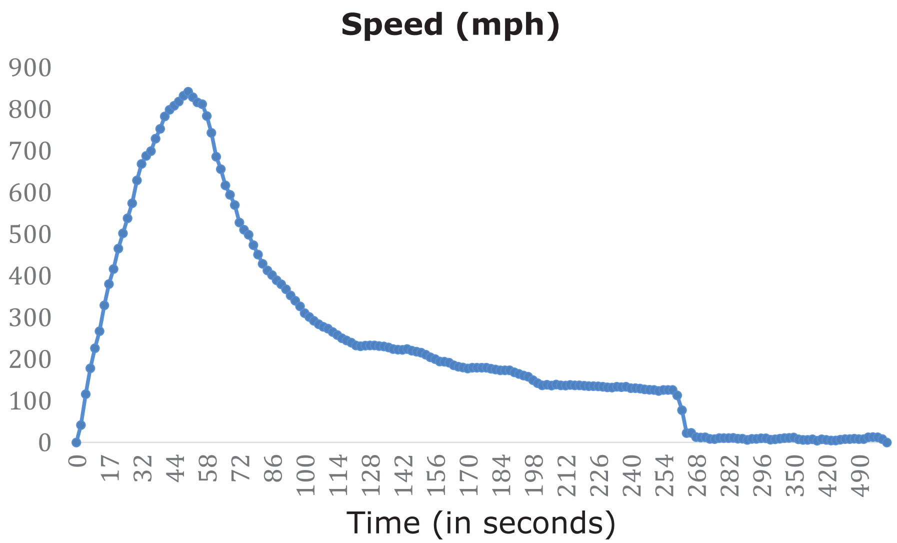
- ☐ I can solve a quadratic using different methods.
- ☐ I can choose a method for solving a quadratic based on the context.

9.10.1 Warm-Up: Free Fall

Free fall is a special type of motion in which the only force acting upon an object is gravity. On October 14, 2012, Felix Baumgartner was the first person to break the sound barrier in free fall when he completed a parachute jump from a height of $127{,}852$ feet (ft).

1. Determine how many miles Felix was above the earth when he jumped out of the capsule. 1 mile $= 5{,}280$ ft

His fastest speed during the free fall was 843.6 miles per hour (mph). Felix spent 4 minutes 22 seconds in free fall and the total jump time was 9 minutes 9 seconds. The graph shows his speed, in miles per hour, during the jump.



2. What do you think happened in Felix's fall at the highest point on the graph?

9.10.2 Collaboration: Different Methods of Solving Quadratics

1. Use the quadratic formula to solve $9x^2 + 36x + 36 = 0$.

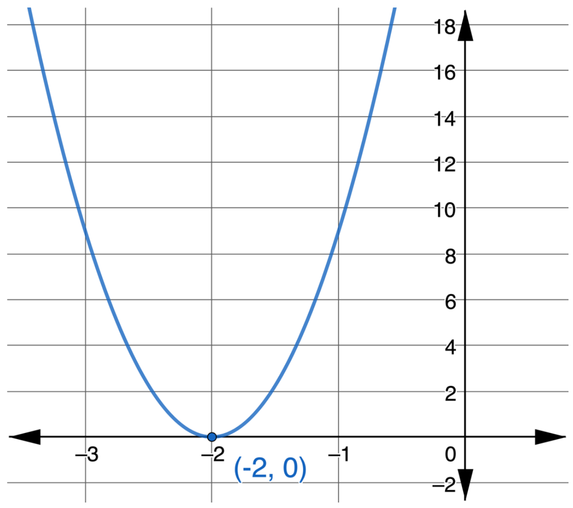
2. Solve $9x^2 + 36x + 36 = 0$ by factoring.

3. Solve $9(x + 2)^2 = 0$.

4. Check your work with your partner and resolve any differences if necessary.

5. The graph of $y = 9(x + 2)^2$ is shown.

   Work with your partner to write an explanation that connects the graph of $y = 9(x + 2)^2$ to the solutions to $9x^2 + 36x + 36 = 0$ and $9(x + 2)^2 = 0$.



6. Work with your partner to complete the statements using the word bank provided.

different	factored	factored form	same
standard form	vertex form	x-intercept(s)	
y-intercepts	zero product property		

The three methods for solving quadratics use different forms of a quadratic. Each method results in the ___________ solution(s). To solve by factoring, the quadratic is factored, then the ___________________________ is applied to find the solutions. To solve by completing the square, the quadratic is rewritten to ___________________________. Then, the equation is solved for x. The quadratic formula is best used when the quadratic cannot be ___________ or if completing the square is difficult. The solution(s) are the _____________ of the parabola. These are also known as the zeros of the graph or roots of the quadratic.

The **roots** or **zeros** of a quadratic equation are the solution(s) of an equation that has the form $ax^2 + bx + c = 0$.

9.10.3 Guided Instruction: Choosing a Method

1. The equation $y = -16x^2 + 128x + 1$ models the height, in feet, after x seconds, of a toy rocket that is launched from 1 foot above the ground.

   a. What is the height of the rocket when it is on the ground?

   b. Use the table to determine when the rocket reaches its maximum height.

Time (seconds)	-2	1	3	5	7	10
Height (feet)	-319	113	241	241	113	-319

   c. The equation $-16x^2 + 128x + 1 = 221$ can be used to determine the time that the rocket was at 221 feet. Rewrite the equation by filling in the blanks.

$$\_\_\_\_\_x^2 + \_\_\_\_\_x + \_\_\_\_\_ = 0$$

d. Consider the three methods of solving quadratics and choose one method to solve the equation.

- ○ Completing the square
- ○ Factoring
- ○ Quadratic formula

Explain your choice.

e. Ask your partner which method they selected. Record their choice and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Method	My Summary of Their Reason	Initials
○ Completing the square ○ Factoring ○ Quadratic formula		

f. Solve $-16x^2+128x+1=221$ showing your work.

g. Complete the statement about the solution to $-16x^2+128x+1=221$.

At ______ seconds and ______ seconds, the toy rocket is at a height of 221 feet.

2. The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, that U.S. passengers traveled on a train for the period 1970 to 2000, where x is the number of years since 1970.

   a. Use the equation $5x^2 - 160x + 6179 = 8279$ to determine in what year U.S. passengers traveled $8{,}279$ millions of miles.

   b. Interpret the solutions and determine if the solutions are viable.

9.10.4 Guided Practice: Choosing a Method

1. The function $f(x) = -3x^2 + 54x + 83$ models the sales of music cassette tapes for the years 1980 to 2000, in tens of millions of dollars, where x is the number of years since 1980.

   a. Use the equation $-3x^2 + 54x + 83 = 0$ to determine when the sales of music cassette tapes are $\$0$.

   b. Interpret the solutions and determine if the solutions are viable.

9.10.5 Wrap-Up: Ticket Out the Door

The function $S(x) = -0.5x^2 + 50x - 8$, where x is time in seconds (s), models the speed, S, in kilometers per hour (km/h) of Felix Baumgartner during the first 90 seconds of his record-breaking free fall.

1. Use the function, $S(x)$, to determine when Felix reached the greatest speed.

2. Mathematical models use data to determine an equation that closely fits but may not be exact. According to the actual data collected during Felix's fall, the greatest speed he reached was 1355 km/h, which occurred at 50 seconds. How well did the function $S(x)$ match the real data?

Unit 9, Lesson 11: Writing and Solving Quadratic Equations

Benchmarks

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non- viable options.

Learning Target

☐ I can write and solve one-variable quadratic equations.